BAV
Recall Sheets

Forty sheets. One topic each.

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XLRI Jamshedpur · PGDM BMJ 2025–27 · Term IV

red *= a trap he has set · **green** = the consistency check · grey = a number worth carrying in*

*These are for the **second and third** pass, not the first.*

Read the Field Manual once, then live here.

The spine

read this first

```
Value = Σ (cash flow to a claim) ÷ (1 + rate matching that claim's risk)^t
```

*FCF · FCFE · CCF · APV · DDM · EVA · multiples · the VC method — all the same equation, different definition of "cash flow to a claim". **They must all give the same answer.***

Four doctrines he grades on

1. **Consistency beats accuracy.** "Justify the assumptions rather than trying to prove they are correct."
2. **Growth is never free.** g = ROIC × Investment Rate. Always.
3. **Count each asset & each claim exactly once.** Never twice, never zero times.
4. **The multiple is an output**, never the starting point.

The four-question audit — run on every answer

1. Does my rate match my cash flow? *(risk)*
2. Every asset counted once? *(asset)*
3. Every claim counted once, same way in flow and bridge? *(liability)*
4. Growth backed by reinvestment, implied multiple sane? *(assumption + multiplier)*

*Half his hard questions are **inconsistent by construction**. The mark is for spotting it, not for calculating.*

The statements

The P&L ladder — memorise

```
Revenue
 − materials, employee cost, other opex
= EBITDA
 − depreciation & amortisation
= EBIT  (= net operating income)
 − interest         ← everything ABOVE belongs to ALL capital
= PBT                 everything BELOW belongs to EQUITY only
 − tax
= PAT  (= net income = the E in P/E)
```

Balance sheet — the distinction that matters

Not current vs non-current. ***Interest-bearing vs non-interest-bearing.***

IB → goes into WACC, gets subtracted at the bridge.

NIB → sits inside working capital, not subtracted.

Depreciation

- *Non-cash. The cash left when the machine was bought.*
- *A convention, not a measurement — real value falls fastest early.*
- ***So accounting ROCE drifts up over an asset's life for no real reason. Don't over-read the trend.***

```
Enterprise value = Debt (MV) + Equity (MV)
Equity value     = EV − debt (+ non-op assets, − other claims)
```

Time value + the two shortcuts


```
PV = CF_t ÷ (1+r)^t


Level perpetuity     PV = C ÷ r
Growing perpetuity   PV = C₁ ÷ (r − g)      needs r > g, always
```

Two traps inside Gordon

1. *Numerator is **NEXT year's** flow. Given this year's → grow it:* $C_1 = C_0(1+g)$.
2. *$g \geq r$ is not "a high valuation" — it is an **impossible assumption set**.*

Terminal value — the two-step both students get half right

```
TV at end of yr N = FCF_N × (1+g) ÷ (WACC − g)
PV of it          = TV_N ÷ (1+WACC)^N       ← N periods, not N+1
```

Getting one of these right and the other wrong is the single most common arithmetic slip in the class.

Implied growth — the rearrangement (Quiz 3 engine)

```
g = (TV × WACC − FCF) ÷ (TV + FCF)
```

Real vs nominal — the tomato

```
(1+nominal) = (1+real)(1+inflation)
3% real, 6% inflation → 9.18%,  NOT 9%
terminal g: 5% real × 5% inflation = 10.25%, NOT 10%
```

Risk, CAPM, and the crack in it

$$\sigma_i^2 = \beta^2 \sigma_m^2 + \sigma_\varepsilon^2$$

total = systematic + specific (diversifiable)

A diversified investor kills specific risk for free ⟹ the market won't pay for it ⟹ only systematic risk is priced. That is CAPM.

$$K_e = R_f + \beta \times (E[R_m] - R_f)$$

$\beta = 1$ moves with market · $\beta = 1.7$ amplifies · $\beta = 0.5$ defensive

The crack — remember it now, use it in Part V

CAPM prices only systematic risk **because it assumes the investor is diversified.**
Remove that assumption — a promoter with 90% of her wealth in one private business —
and CAPM **understates** what she needs.
⟶ that single crack is worth a 6× swing in the Kavita case.

His own caveats on CAPM

- Fama–French (1992): beta doesn't explain returns.
- Low-risk stocks outperform (Martingale AM).
- Alternatives: APT (must estimate factor premia), Fama–French multi-factor (industry standard, IIMA publishes Indian data).

The five consistencies

★ *highest yield*

#	Rule	Violation he circles
1	**Assumptions** — *projections tell one story; growth needs reinvestment*	*15% sales growth, flat capex & NWC. Terminal g = 3% with NI = 0.*
2	**Risk** — *rate matches the flow*	*FCFE at WACC. Growing firm, constant WACC, debt fixed in ₹.*
3	**Asset** — *each asset once*	*Interest income out of NOPAT but cash not added back. Adding land whose output is in revenue.*
4	**Liability** — *each claim once, same way in flow and bridge*	*Liability in WACC but not subtracted from EV.*
5	**Multiplier** — *the multiple is an output*	*Terminal g = 10% next to terminal EV/EBITDA of 4×.*

> *"Name the consistency violated" = the cheapest 5 marks in the paper.*
> *Practise writing all five names in 30 seconds.*

Absolute vs relative

DCF = *compute it yourself. "The only scientific method, in the absence of real options."*
Multiplier = *borrow the market's opinion. Valid only if the comparables are fairly priced — and it cannot see a whole sector being wrong.*
Historical cost = *what it cost to build. Useful for insurance, useless for value.*

The growth engine ★

```
g = b × ROE          b = retention ratio = 1 − payout
g = ROIC × Investment Rate      IR = Net investment ÷ NOPAT
```

Retain nothing → grow 0%. Retain everything → grow at ROE. No third option.

> *His words (Quiz 3 key): changing g directly is* ***incorrect*** *— "it is an input to the model".*
> *Given g, ROIC and NI that don't satisfy the identity ⟹* ***one of them is wrong.***
> *Back out the consistent set.*

NPVGO — growth only pays above the cost of capital

```
Value = EPS₁ ÷ K_e  +  NPVGO
ROE = K_e  ⟹  NPVGO = 0  ⟹  growth creates nothing
```

Where to invest? (his slide) — OIC ₹100m, WACC 15%, g 5%

Co	ROIC	Spread	Verdict
A	10%	−5%	**growth DESTROYS value; worth < 100**
B	15%	0	neutral; worth exactly 100
C	20%	+5%	**growth creates value; worth > 100**

It's the <u>spread</u>, not the growth rate. 2% growth on a 10-pt spread beats 20% growth on a negative one.

Modigliani–Miller

Slice a pizza into 6 instead of 4 — no more pizza. Debt and equity are slices of the same cash flows.

```
No tax:   V_L = V_U                        (Prop I)
          K_e = ρ + (ρ − K_d)(D/E)          (Prop II)

With tax: V_L = V_U + t·D                   permanent FIXED debt only
          K_e = ρ + (ρ − K_d)(1−t)(D/E)
```

ρ (rho) *= unlevered cost of equity = pure business risk = what shareholders would demand with no debt at all.*

Quiz 1 Q4 — worked, with his mark scheme

Unlevered, EBIT 200 forever, K_e 12%, t 20%. Borrows 50, invests in a project with NPV 10.

```
V_U   = 200(0.8) ÷ 0.12 = 1,333.33
+ tax shield  = 50 × 0.20 =    10.00
+ CASH RAISED, still in firm =  50.00   ← the one everyone forgets
+ NPV of project            =  10.00
= new EV                      1,403.33   3 marks
                              1,353.33   2 marks (forgot the cash)
                              1,343.33   1.5 marks (forgot cash + NPV)
```

> *He pre-maps the wrong answers ⟹* ***show every line of the build-up.***
> *A visible structure with a slip earns most marks; a bare wrong number earns none.*

The value-driver formula

★★ *generates everything*

$$P_0 = E_1(1 - g/ROE) \div (K_e - g)$$

$$P/E = (1 - g/ROE) \div (K_e - g)$$

$$P/B = (ROE - g) \div (K_e - g)$$

Set g = 0 and read off the exam answers

$P/E \rightarrow 1 \div K_e$ ROE term VANISHES $\Longrightarrow$ P/E says NOTHING about ROE

$P/B \rightarrow ROE \div K_e$ $\Longrightarrow P/B > 1 \iff ROE > K_e$

Quiz 1 Q1: *at $g = 0$, which ratio reveals ROE > required return?* → ***P/B.***
1 mark for saying P/B, 1 mark for the justification. Bare answer caps at half.

Quiz 1 Q2: *lower P/E and higher P/B?* → ***Yes: low g + high ROE.*** *A mature, cash-generative, high-return business with no reinvestment runway. 3 marks.*

Only three determinants of any multiple

growth (g) · return quality (ROE / ROIC) · risk (K_e / WACC)
Two companies at different multiples ⟹ one of these three differs, or the market is wrong about one.

P/E, PEG, PEGY

Quiz 1

P/E hygiene

- *Always **diluted** — convertibles, warrants, ESOPs are real claims.*
- *Forward > trailing for value, but imports someone else's estimate.*
- ***Normalise E:** strip asset-sale gains, insurance, restructuring, litigation, mark-to-market.*

> *Alphabet Q2 2026: reported diluted EPS **$9.11** included ~$99bn of unrealised gains on SpaceX + Anthropic stakes. Adjusted EPS **$2.85** — which MISSED the $2.89 consensus. A P/E on 9.11 is meaningless.*

PEG

```
PEG  = (P/E) ÷ g           g as a number: 20% → 20
PEGY = (P/E) ÷ (g + dividend yield)
```

Why he's sceptical: P/E is convex in g (g sits in numerator via payout AND denominator via K_e − g), so PEG flatters high growth. And it ignores risk entirely — same PEG, different betas, not equally attractive.

DMart: 75× FY22 P/E, 26% CAGR ⟹ PEG 4.4×. Defended because Walmart 5.3×, Costco 4.9×, Target 7.3×. Notice the move: a rich multiple is defended by choosing the comparison set.

P/B, P/S, and multiplier consistency

P/B

```
P/B = (ROE − g) ÷ (K_e − g)   →  at g=0: ROE ÷ K_e
```

- P/B = 1 → earns exactly K_e, creates nothing
- P/B < 1 → market expects ROE < K_e (PSU banks, struggling capital-heavy firms)
- P/B > 1 → ROE > K_e; the gap reads the spread

Use it for balance-sheet businesses: banks, NBFCs, insurers, oil & gas, real estate — paired with ROE. Useless for asset-light IT services.

P/S — Quiz 1 Q3, his two reasons

1. **Leverage.** Price is an EQUITY claim; sales belong to the WHOLE firm. Same operations, different debt → different P/S. 1 mark
2. **Cash.** A cash pile lifts market cap without a rupee more revenue. 1 mark

The general rule, worth more than the answer:
numerator and denominator must belong to the same claimants.
equity ↔ equity (P/EPS, P/BV) · firm ↔ firm (EV/EBITDA, EV/Sales, EV/IC)
EV/Sales is the fixed version of P/S.

EV/EBITDA + choosing peers

Quiz 1 / practical

```
EV/EBITDA = (Mkt cap + Debt − Cash) ÷ EBITDA
```

Virtues: *capital-structure neutral, depreciation-policy neutral, rough cash proxy. The multiple most deals are struck on.*

Defects: ignores capital intensity entirely (a steel plant and a software firm with equal EBITDA are not equally valuable); ignores working capital; and Ind AS 116 broke comparability — rent left opex, so EBITDA jumped for retail / aviation / cinemas with zero change in economics.

EBITTDA — 2026 (Quiz 3 Q3)

> **For:** in early AI adoption, token spend is exploratory → closer to **R&D** than to COGS.
>
> **Against:** if AI is embedded, revenue is already higher — excluding the cost breaks the **matching concept**; and it **encourages profligate token usage.**
>
> *one each way = 2 of 3 marks. Label them "For:" and "Against:".*

Peers & adjustments

Screen on **business model, growth, margin, risk, size** — not SIC/GICS codes.

Premium: scalability, pricing power, intangibles, high ROCE, fixed-asset turnover, short NWC days, cash conversion.

Discount: margin contraction, lagging growth, client concentration, promoter dependency, sunset industry.

Quiz 1 decoded

past paper

Q	Asked	Mk	Testing
1	$g = 0$: P/E or P/B for ROE > r?	2	P/E collapses to $1/K_e$
2	Lower P/E + higher P/B possible?	3	low g, high ROE
3	2 reasons not to use P/S	2	multiplier consistency
4	Unlevered firm borrows & invests → EV?	3	MM + the cash stays in the firm

> Pattern: **3 conceptual, 1 numerical** — and the conceptual questions carry MORE total marks. Answers are 2–3 sentences. He tests whether you can name the mechanism, not write an essay.

How he marks — calibrate before writing

- **Justification = half the marks.** "1 mark for saying PB, 1 for the justification."
- **Partial credit is pre-mapped** ⟹ show the build-up.
- **Structure earns marks** on argue-both-sides questions.
- **Conceptual answers are SHORT.** Causal chain, then stop.
- **He concedes his own errors** ("Give 1 mark to all. This was a mistake.") ⟹ **attempt everything.**
- "Ignore the circularity" means he stripped it. If he doesn't say it — **the circularity IS the point.**

Accrual → cash

```
EBIT × (1−t)                = NOPAT
 + depreciation & amortisation
                            = Gross cash flow
 − capex
 − increase in operating working capital
 (+ increase in other NIBL)
                            = FREE CASH FLOW
```

Why tax EBIT, not PBT? *FCF must be pre-financing. Taxing PBT sneaks the interest deduction into the numerator — and WACC's (1−t) already counts it. Double count.*

Why is depreciation ALONE added back? (Quiz 3 Q2, 5 marks)

NOT "because it's non-cash" — that's half an answer.

Capex ≡ Δ Net fixed assets + Depreciation

Depreciation is already inside the capex you are about to subtract.

*Adding it back to NOPAT **cancels** that. The two moves are a matched pair.*

Operating working capital

```
OWC = operating current assets − operating current liabilities
      (inventory + receivables       (payables + other CL
       + other receivables,           EXCLUDING short-term
       EXCLUDING surplus cash)         borrowings − they are DEBT)
```

Titan FY23: current liabilities ₹11,521 cr → operating CL ₹5,018 cr once ₹6,503 cr of borrowings comes out.

FCF · FCFE · CCF

★ Quiz 2

The question that fixes everything: whose cash is this?

Flow	Whose	Discount at	Gives
FCF	*everyone*	WACC	EV
FCFE	*shareholders*	K_e	Equity directly
CCF	*everyone + the shield*	**PRE-TAX WACC**	EV

```
FCFE = PAT − net investment + net new borrowing
CCF  = FCF + t·K_d·D

WACC     = (E/V)K_e + (D/V)K_d(1−t)      ← pairs with FCF
Pre-tax  = (E/V)K_e + (D/V)K_d           ← pairs with CCF, NO (1−t)
```

The shield goes in the numerator (CCF) **or** *in the rate (WACC). Never both, never neither.*
Using after-tax K_d with CCF $\Longrightarrow$ EV too high. This is the intended failure in Quiz 2.

The reconciliation identity — your diagnostic

```
FCFE = FCF − Interest(1−t) + Net new borrowing
```

9 times out of 10 the mismatch is: constant WACC ($\Longrightarrow$ constant D/E) while debt held fixed in ₹.
Constant D/E + growth $\Longrightarrow$ ***net new borrowing = g × D every year.***

OLP Limited — the benchmark

★ memorise

X = ₹100m forever · t = 20% · D = ₹100m @ K_d 10% · K_e = 20% · no non-op assets

Method	Flow	Rate	Value
FCFE	(100−10)0.8 = 72	K_e 20%	E = 360
FCF	100 × 0.8 = 80	WACC 17.391%	EV = 460
CCF	80 + 0.2(10) = 82	pre-tax 17.826%	EV = 460
APV	80 at ρ, shield apart	ρ = 18.182%, VTS = D·t = 20	440+20 = 460

Equity = **360** under all four. EV = **460** under all four.
Notice the ORDER: FCFE first (needs no weights) → equity 360 → only THEN can you build the WACC weights.
That ordering is the escape from circularity.

WACC weights — why market value

Titan: book ₹230/share, trades at ₹4,629, K_e = 12%. **12% of what?** An investor who can sell at 4,629 expects 12% on 4,629.

His question: X invested ₹100 in 2002; book ₹300, market ₹1,000. Why care about market value? **Because ₹1,000 is what you give up by not selling — the opportunity cost.**

Quiz 2 Q1 — worked

past paper

Secured 100 @7% · unsecured 50 @8% · K_e 12% · EBIT 200 · t 30% · g 4% · NI 25

```
— FCFE road —
Interest  = 100(7%) + 50(8%)                    =   11
PBT = 189 → tax 56.7 → PAT = 132.3
FCFE = 132.3 − 25 + (150 × 4%)                  =  113.3
Equity = 113.3 ÷ (12% − 4%)                     = 1,416.25   2 mk
EV = 1,416.25 + 150                             = 1,566.25

— CCF road —
CCF = 113.3 + 11 − 6                            =  118.30    2 mk
pre-tax WACC = (100/1566.25)7% + (50/1566.25)8%
             + (1416.25/1566.25)12%             = 11.553%   2 mk
EV = 118.30 ÷ (11.553% − 4%)                    = 1,566.25   1 mk
```

① *pre-tax WACC uses the RAW 7% and 8% — no (1−t). After-tax here is the intended failure.*
② *both debts appear separately at their own costs — do not average first.*
③ *net new borrowing = g × total debt = 4% × 150 = 6. **+** in FCFE, **−** going back to CCF.*

Q2 — 5 marks, three sentences

+NPV project ⟹ equity value ↑ by the NPV; debt unchanged ⟹ ***D/E falls*** *⟹ K_e falls (MM II) and weights shift to the cheaper source ⟹* ***WACC falls.***

Growth, and the hidden clause


```
Net investment = capex + ΔWC − depreciation
FCF = NOPAT − net investment
```

The clause hidden in "constant WACC":
constant WACC ⟹ constant D/E ⟹ if equity grows at g, **DEBT MUST GROW AT g**
⟹ net new borrowing of g×D appears in FCFE every year.

Growing FCFE at a constant K_e with debt fixed in ₹ **cannot be right** *— leverage would fall every year, so K_e would fall every year.*
Either (a) let debt grow at g, or (b) hold debt fixed and use APV.

Terminal value

- *g bounded by* **nominal GDP** *of the economy it depends on. Build it multiplicatively.*
- *Explicit period runs* **until steady state is credible** *— longer if ROIC ≫ WACC. The choice is examinable.*
- **Cross-check both ways:** *what EV/EBITDA does my Gordon TV imply? what g does the peer multiple imply?*
- *Report TV as % of EV. Always sensitise WACC × g.*

Terminal reinvestment must comply

```
IR = g ÷ RONIC     Net investment = NOPAT × IR
```

Circularity

WACC needs market-value weights. Equity value is what you're solving for. The snake eats its tail.

	Response	Verdict
1	Book-value weights	**WRONG. Never silently.**
2	Current market cap	Defensible if disclosed — but you're using the price you claim to be testing.
3	**Target D/E**	**Only the RATIO is needed ⟹ circularity dissolves. Best when migrating to a target.**
4	**Iterate** (his SSRN 413240)	**Seed → WACC → value → implied equity → feed back. Excel: iterative calc, or Goal Seek to zero.**

> **The elegant escape:** value equity DIRECTLY with FCFE at K_e. No weights ⟹ no circularity at all.
> That is exactly why his Quiz 2 key computed equity (1,416.25) first.
> Multiple debts at different rates + "find EV"? → FCFE road is nearly always fastest.

XYZ — three legitimate routes, one answer

₹100m @12%, ₹200m @14%, EBIT 100, t 30%, K_e 20%, $g = 0$

```
FCFE: interest 40 → PAT = 60(0.7) = 42 → Equity = 42/0.20 = 210
one debt as capital → different WACC, different EV — same 210
both debts as capital → EV 510, subtract 300 — same 210
```

APV + preference capital


```
APV = V_unlevered + PV(tax shields) − PV(distress costs)

Step 1   find ρ:   K_e = ρ + (ρ − K_d)(1−t)(D/E)    → solve
         or        β_U = β_L ÷ [1 + (1−t)(D/E)]
Step 2   discount FCF at ρ
Step 3   value the shields — rate depends on the DEBT POLICY
```

Debt policy	Shield rate	Unlever with
Fixed in ₹, known schedule	K_d (zero-g: VTS = $D \cdot t$)	$\beta_L \cdot E / [D(1-t)+E]$
Constant D/V, continuous (Harris–Pringle)	ρ	$\beta_L \cdot E / (D+E)$
Constant D/V, annual (Miles–Ezzell)	yr 1 at K_d, then ρ	M–E adjustment

> **From class (APV lecture):** "Mistake — using the MM equation, based on the assumption $g = 0$. When you bring growth in, the riskiness of the tax shield changes."
> Debt grows with the firm $\Longrightarrow$ **drop the (1−t)** from relevering; discount the shield at ρ, not K_d.

Preference capital — leverage without the tax break

```
WACC = (E/V)K_e + (D/V)K_d(1−t) + (P/V)K_p     ← no (1−t) on K_p
and SUBTRACT preference at the bridge.
```

Use APV whenever leverage is CHANGING — LBO, deleveraging plan. One constant WACC is then simply wrong. (Sampa Video: fixed debt → APV at K_d; constant D/V → WACC.)

Four technical points + the −5.12

★ Quiz 3

1. **Pre-tax cost of debt ≠ coupon rate.** It's the YTM.
2. **Cost of equity ≠ dividend yield.** Total return = yield + capital gains.
3. **Weights at market value**, not book.
4. **Rate must match the riskiness AND the type** of the flow.

Does the coupon-vs-YTM error matter?

EBIT 100, book debt 100, t 30%, K_e 20%, YTM 12%, coupon 10%, g = 0

	coupon	YTM
MV of debt	100	83.33
WACC	16.87%	17.57%
EV	415.00	398.33
Equity	**315**	**315**

At g = 0 it does NOT matter — equity is 315 either way. The problem arrives with a decision.

The ₹500m project at 17% IRR — clears 16.87%, fails 17.57%.

Extra debt (100/415)500 = 120.48 · extra equity 379.52

Project FCF 85 → EBIT 121.43 → new EBIT 221.43

New interest 10 + 120.48(12%) = 24.46 → PBT 196.97 → PAT 137.88

New equity = 137.88/0.20 = 689.40

Put in 315 + 379.52 = 694.52 ⟹ **NPV = -5.12**

The 17% project DESTROYED value ⟹ 17.57% was the true cost of capital.

Cost of debt

The pecking order

1. **YTM of the company's traded bonds.** Best — and rare: only 83 of BSE 500 borrowed from the public (FY23).
2. **Ask someone who knows** (Bloomberg, treasury).
3. **Credit rating + spread.** FIMMDA publishes annualised spreads by rating × tenor. The practical route.
4. **Rate the bonds yourself** — synthetic rating from coverage & leverage.

FIMMDA shape: AAA a few bp short-tenor → ~117bp at 15y; BBB− 450–560bp. Spreads widen with tenor, explode below investment grade.

Which tax rate in K_d(1−t)?

Marginal statutory. "What does the next rupee of interest save?"

§115BAA = **25.168%** (22 + 10% surcharge + 4% cess) · old regime ≈ 34.94% · §115BAB ≈ 17.16%

Verify from the AR tax note. Never assume.

Foreign currency loans

US$10,000 @ 6%, spot ₹64/$, ₹ rate 7%, $ rate 1%

Expected spot rises 64 → 85.41 over 5 years

Implied rupee cost of debt = 12.30%, not 6%.

The extra 6 points = expected rupee depreciation.

```
IRP:  Forward = Spot × (1 + r_₹) ÷ (1 + r_$)
      80 × 1.07/1.05 = ₹81.52/$
```

★ Liability consistency

★★ highest yield

	D_1 IS a source of capital	D_0 is NOT
WACC	include, at its own cost	exclude
Cash flow	do NOT subtract its interest	DO subtract its interest; ADD any increase in it
Bridge	SUBTRACT from EV	do NOT subtract

> Both columns are correct. **Mixing them is never correct.**

Are non-interest-bearing liabilities really free?

Buy raw material on 1 year's credit, pay ₹1,100 cr; K_d = 10%. Cash price would have been ~₹1,000 cr. **The ₹100 cr difference is interest in disguise** — the supplier priced it into the invoice.

Convention keeps payables in working capital, so the finance cost is buried in COGS. Say this if asked — the marks are for seeing that "non-interest-bearing" is a label, not a fact.

Debt-like items — the "what else do you subtract?" list

> borrowings · **unfunded gratuity & leave encashment** · **stretch creditors** (60-day terms run at 180 = 120 days of disguised borrowing) · non-convertible preference shares · lease liabilities · contingent liabilities (impact × probability) · minority interest

Cash: the two treatments

★ *Quiz 3*

Zero growth · EBIT 100 (excl. interest income) · debt 100 @10% · cash 50 @7% · t 30% · K_e 20%

ANCHOR — value equity directly, it cannot depend on the choice:
PBT = 100 + 3.5 − 10 = 93.5 → NI = 65.45 → **Equity = 327.25**

Cash = OPERATING

FCF = (100+3.5)0.7 = 72.45

EV = 327.25 + 100 = 427.25

WACC = 16.957%

check 72.45/16.957% ✓

Cash = NON-OPERATING

FCF = 100 × 0.7 = 70

EV = 327.25+100−50 = 377.25

WACC = 18.55%

(effective K_d on NET debt = 13%)

check 70/18.55% ✓

Two EVs, two WACCs, ***ONE equity value: 327.25.***
EV is not a fact about the world — it's a fact about your definition of "the enterprise".
Equity value is the invariant. *If your two treatments disagree, something is broken.*

Deciding: operating cash = the working balance the business genuinely needs. Surplus cash & marketable securities = non-operating by default, especially when large. Whichever you pick — same choice in the flow, in the beta, and in the bridge.

CAPM inputs: R_f and the ERP

Risk-free rate

- 10-year G-Sec (FBIL/RBI), as of the valuation date. Cite the print date.
- **Which maturity?** CAPM is a one-period model, but we value perpetuities → long bond is least-bad.
- **Really risk-free?** India has a sovereign CDS spread. Nominally default-free in ₹, not inflation-free.

> **Yield curve not flat** — CFs 100,100,200; spots 5%,6%,7%
>
> `Value = 347.50 → solve →` **`r = 6.52%`**
>
> `front-loaded (100,10,10)     → 5.49%`
>
> `back-loaded (100,100,10000) → 6.99%  ≈ the 3-yr spot`
>
> *A DCF is dominated by its TV ⟹ anchor at the long end.*

Market risk premium — nobody publishes it

- **Historical:** index return − R_f over a long window. Sensitive to window & arithmetic-vs-geometric.
- **Implied (Damodaran):** solve for the rate that equates today's index level with consensus cash flows. Forward-looking. India ≈ **7.08%**.
- **Convention: 6–7%** in Indian practice.

> Market portfolio must be **broad** and **VALUE-WEIGHTED**.
> Most indices are PRICE indices ⟹ they exclude dividends ⟹ **add the dividend yield back** or you understate the premium by the whole yield.

Beta, start to finish

★ **Quiz 3**

$$\beta = Cov(r_i, r_m) \div Var(r_m) = \rho \times (\sigma_i \div \sigma_m)$$

1. Returns

*Use **adjusted** prices. Adjust for dividends, bonus, splits, **rights**.*

*RIL 2021, 1:15 rights at ₹1,257, price ₹1,435 → theoretical ex-rights = (15×1435 + 1257)/16 = **₹1,423.87**. The "0.8% fall" is an artefact, not a return.*

2. Period & frequency — two principles in tension

- *Longer = statistically better (tighter SEs)*
- ***BUT operating and financial risk must not have changed over the window***

***Daily:** ~1,300 points over 5y, but **infrequent trading** (no trade Tuesday = fictitious return) and **non-synchronous trading** (positive serial correlation, beta biased DOWN). Thin stock → weekly/monthly.*

> *Tata Steel: β goes **1.348** → **1.717** just by switching to 60 monthly returns.*
> *The number is a fact about the estimation choice. **Disclose the choice.***

3. Unlever → median → relever

$$\beta_U = \beta_L \div [1 + (1-t)(D/E)] \qquad \beta_L = \beta_U \times [1 + (1-t)(D/E)]$$

*4–6 peers · unlever each with **its own** D/E · take the **MEDIAN** · relever at target D/E. **Net debt** usually beats gross. Net D/E can be **negative** for a cash-rich firm ⟹ $\beta_U > \beta_L$.*

WACC assembly + the bridge

Seven steps

1. List every claim; mark IB / not, source-of-capital / not. **Write it down** — needed twice more.
2. K_d: YTM or rating+spread; after tax at marginal statutory.
3. K_e: CAPM. R_f, ERP as a stated assumption, β regressed + bottom-up cross-check.
4. Other claims (preference, leases, FX loans) at their own costs.
5. Weights at market value; circularity by target D/E or iteration — **say which**.
6. Blend.
7. Sanity: is WACC between K_d(1−t) and K_e? plausible for the sector? comfortably above terminal g?

The bridge

```
  PV(explicit FCF) + PV(terminal value)      = EV of operations
+ non-operating assets   surplus cash, financial investments, vacant land,
                         assets held for sale, non-op advances to related
parties
- debt & debt-like items
- contingent liabilities  impact × probability
- minority interest       if consolidated
= EQUITY VALUE
```

Quiz 3 Q4 — 6 of 7 were "NOT NEEDED"

brands ✗ · financial investments whose income is in NOI ✗ · factory land ✗ · cash (gross-debt WACC) ✗ · customer advances ✗

contingent liability ✓ (a % of it) · the debt item — "give 1 mark to all, this was a mistake"

Default answer = no adjustment. Ask: is this asset's income already in my cash flow?

Building FCF (Titan)

Line	Built as
Revenue	net sales, **restated** for disposals/acquisitions (Walmart Intl 121→101bn = sold Asda + Seiyu; Oyo booked 22% commission, not the hotel bill)
Opex	each head as % of revenue, 3–5 yrs (Titan raw materials ~76%; employee cost 4.6→3.6% = operating leverage)
Depreciation	**% of NET FIXED ASSETS — not of sales**
NOPAT	**historical → CASH tax rate; projections → STATUTORY rate**
Gross cash flow	NOPAT + depreciation
Capex	**Δ NFA + depreciation.** Lumpy — link to a revenue driver (stores planned, outsourcing shift)
Gross investment	capex + ΔOWC − ΔNIBL
FCF	gross cash flow − gross investment Titan FY22 = **−3,160 cr** (WC +5,020 cr). Negative in a growth year is fine.

Other operating income — the 2x2

	main business? YES	NO
recurring? YES	OPERATING	non-operating
recurring? NO	IGNORE	IGNORE

retailer's credit-card income → operating · drug retailer selling containers → operating · investment income → depends trade vs non-trade · **gain on sale of assets → IGNORE**

Standalone vs consolidated: same business & geography → consolidated. Different lines / operating environments → standalone + value investments separately (his Tata Steel logic; investments were >35% of value there).

ROIC · EVA · growth

```
OIC  = net fixed assets + operating working capital
ROIC = NOPAT_t ÷ OIC_(t−1)          ← note the LAG
EVA  = OIC_(t−1) × (ROIC_t − WACC_t) = NOPAT − WACC·OIC_(t−1)
EV   = OIC + PV(all future EVA)
g    = ROIC × Investment Rate    IR = net investment ÷ NOPAT
```

EVA and FCF must agree

OIC_0 = 100, ROIC 20%, IR 25%, WACC 13%, terminal g 5%

Yr	1	2	3	4 (T)
OIC (open)	100	113	125	135
NOPAT	20.0	22.6	25.0	27.0
Net inv	13.0	12.0	10.0	6.75
FCF	7.0	10.6	15.0	20.25
EVA	7.00	7.91	8.75	9.45

> Both give **200.32**. Terminal NI = (g/ROIC) × NOPAT = (5/20) × 27 = **6.75**. Leave NI at 10 in the terminal year and they split: EVA 198.05, FCF 161.34. **That divergence is the alarm bell.**

> From class: "**EVA method estimates for t+1 years** — restate year 4 numbers." EVA_t needs OIC_(t−1) ⟹ an N-year model needs the N+1 forecast.

★ The terminal-year engine

★★ Quiz 3 Q1

*Maura Lartin's terminal year: NOPAT \$100m · **NI = \$0** · EBITDA multiple 9× · WACC 10% · EBITDA \$130m · ROIC 12%*

> ***The mistake:*** *net investment cannot be zero when terminal g is positive.*
> *g = ROIC × IR, so IR = 0 forces g = 0 — but her own TV implies positive g.* 2 marks

```
Her FCF = 100 − 0          = 100              1 mk
Her TV  = 9 × 130          = 1,170            1 mk

g    = (1170×0.10 − 100) ÷ (1170 + 100)
     = 17 ÷ 1270           = 1.3386%          1 mk
IR   = g ÷ ROIC = 1.3386 ÷ 12     = 11.155%   1 mk
NI   = 100 × 11.155%       = 11.155           2 mk
FCF  = 100 − 11.155        = 88.845           1 mk
TV   = 88.845 × 1.013386 ÷ (0.10 − 0.013386) = 1,039.49   1 mk
```

The naive answer overstates TV by ₹130m ≈ 12.5% — it credits growth the company never paid for.

The generalised engine — run on ANY terminal year

1. Write down every terminal quantity given.
2. Test g = ROIC × (NI ÷ NOPAT). Fails? one input is wrong.
3. Pick the most reliable **anchor** (usually the exit multiple or WACC — they come from the market).
4. Back out implied g = (TV·WACC − FCF) ÷ (TV + FCF).
5. Force the rest: IR = g/ROIC · NI = NOPAT×IR · FCF = NOPAT − NI.
6. Recompute TV with Gordon.

The VC method

1 Terminal value = exit-year earnings × comparable multiple

2 FV of investment = Investment × (1+r)^n

3 **Required % = FV of investment ÷ Terminal value**

4 Post-money = Investment ÷ required %

5 Pre-money = Post-money − Investment

6 New shares = Old × [% ÷ (1−%)]   Price = Investment ÷ New shares

Cross-check every time: **price per share = pre-money ÷ OLD shares.**

NewVenture — raise $5m, NI $5m in yr 5, 20x comps, r = 50%, 1m shares

TV = 20 × 5 = **$100m**

FV = 5 × 1.5^5 = 5 × 7.59375 = **$37.97m**

Required = 37.97/100 = **37.97%**

Post = 5/0.3797 = 13.17 · Pre = **8.17**

New shares = 1m × (0.3797/0.6203) = **612,100** → price **$8.17**

The option pool — gross UP, never add

Management must own 15% at exit ⟹ she still needs 37.97% at exit, so today:

37.97% ÷ (1 − 15%) = **44.68%**

founder's 1m = 55.32% → total 1.81m → VC 0.81m → price **$6.19**

*Dilution is **multiplicative**, not additive. 37.97 + 15 = 52.97 is the trap.*

The pool is funded entirely out of the founder's pocket: $8.17 → $6.19.

*Raise $12m instead of $5m → FV 91.13 → VC takes **91.13%**, price collapses to $1.17. Raising more than you need is extraordinarily expensive.*

Two rounds + anti-dilution

Two rounds — work BACKWARDS from the exit pie

R1 $5m @50% for 5y · R2 $3m at start of yr 3 @30% for 3y · mgmt 15% by yr 5

```
shares of the yr-5 pie:
VC2 = 3 × 1.3³ = 6.591 → 6.591%   VC1 → 37.97%   mgmt 15%   founder 40.44%

walk back, one dilution at a time:
VC2 at round 2 = 6.591 ÷ (1−0.15) = 7.75%
VC1 today      = 44.68 ÷ (1−0.0775) = 48.43%

yr 0: total 1 ÷ 0.5157 = 1.939m · VC1 0.939m · price $5.33
yr 2: total 1.939 ÷ 0.9225 = 2.101m · VC2 0.162m · price $18.52
yr 5: total 2.101 ÷ 0.85 = 2.472m · exit price $40.45
```

> ***Exit shares first, then gross up backwards by ÷(1 − dilution).*** *Never work forwards.*

Anti-dilution — A bought 10m @$1.00, founders 15m, now $5m at $15m pre

```
weighted avg:  NCP = [(OB × OCP) + New$] ÷ OA
```

Regime	B price	Founders	Series A	Series B
none	$0.6000	45.0%	30.0%	25.0%
weighted avg	$0.5714	42.9%	32.1%	25.0%
full ratchet	$0.3333	**25.0%**	50.0%	25.0%

> ***Series B always gets 25%.*** *Protection does not take from the NEW investor — it transfers value* ***from the founders to the earlier investor.***

Private companies — the four gaps

1. **Financial data.** Different standards (Ind AS only above ₹250 cr), accounts prepared to minimise tax, not to inform.
2. **No market price** ⟹ no market cap, no return series, **no beta**, no MV weights.
3. **Concentrated ownership** ⟹ owner undiversified ⟹ CAPM's core assumption fails.
4. **Illiquidity.** You can't sell on Tuesday.

+ his fifth: corporate governance tells you about problems that are going to come.

Normalising — Kavita Textiles FY2012

Item	Reported	Restated	Why
Wages	1.20	2.16	owners draw ₹1L/mo; market ₹5L ⟹ +2×4L×12 = 0.96 cr
S&D	1.50	1.00	personal expenses
Admin	1.30	1.00	personal expenses
PAT	6.80	**6.60**	

Checklist: owner comp above OR below market · personal expenses (cars, home office, travel, seminars) · **ghost employees** · non-recurring items · personal assets on the books. Pass-through entities: no entity-level tax ⟹ different after-tax CF and shield.

Beta with no share price

Unlever 5 peers with their own **net** D/E → **median** $\beta_U = 0.52$ → impute MV equity = 17 × ₹6.6 cr = **₹112.71 cr** → relever → $\beta_L = 0.51$ → $K_e = 8.2\% + 0.51(7.13\%) =$ **11.84%**

The undiversified owner

★★ *interim*

*CAPM prices only systematic risk. Kavita bears **total** risk. In his words, theory "simply assumes people like Kavita to be irrational."*

Approach	K_e	Equity value
Plain CAPM (diversified)	11.84%	₹459.38 cr
"double the beta"	15.47%	–
***"double the K_e"** (used in case)*	23.67%	**₹79.15 cr**
Total-risk (market model)	71%	–

market model: $r_{i,t} = \alpha + \beta\, r_{m,t} + \varepsilon$

$$\sigma_i^2 = \beta^2\sigma_m^2 + \sigma_\varepsilon^2$$

Total beta = $\beta \div \rho = \beta \div \sqrt{R^2}$

*Non-systematic risk ≈ **5x** systematic in this sector ⟹ ~5x the premium ⟹ $K_e \approx 71\%$.*

ONE judgement call → a 6x swing.

₹459.38 cr vs ₹79.15 cr — before illiquidity and growth are even applied.

*"Biggest source of uncertainty in private valuation?" **The cost of equity.** Not the cash flows, not the DLOM.*

WACC = (2/114.71)(11%)(0.75) + (112.71/114.71)(23.67%) = **23.95%**

PV(FCF) 77.15 − debt 5 + cash 7 = **equity ₹79.15 cr**

DLOM · control · key person

interim

DLOM — discount for lack of marketability

- **Restricted stock studies** — privately placed shares of listed cos: **20–35%**
- **Pre-IPO studies** — placement vs subsequent IPO: **30–50%** (narrower recently)
- **Option pricing (protective put)** — the only model: a function of volatility and holding period

Case uses a flat **25%**: 79.15 × 0.75 = **₹59.36 cr**. Sensitivity: 15% → 67.28 · 30% → 55.41. ≈₹12 cr swing — an order of magnitude smaller than the K_e call.

Control premium / minority discount

> Control premium **20–30%** · minority discount **15–25%**
>
> **A DCF on management's forecasts is a CONTROL value by default** — it assumes you can implement the plan. Valuing a minority stake on it without a discount overstates it.

Key person — the counter-intuitive one

> "Value of the business is **inversely proportional** to the stature of the key person. The buyer buys the company minus the key person."
>
> Mitigations: non-compete · key-man insurance · management depth · **earn-out / escrow tied to retention**

Three transaction types

Private → private: neither side diversified, illiquid, key-person bites → high K_e, full DLOM.

Private → public: buyer's investors ARE diversified → K_e toward CAPM, DLOM shrinks.

Private → IPO: DLOM gone, K_e at plain CAPM, disclosure arrives.

The Kavita value ladder

Step	Equity value	What changed
Market multiple (17× adj. earnings)	112.71	imputed, only for WACC weights
Fully diversified (K_e 11.84%)	459.38	plain CAPM, no growth
No-growth, undiversified + illiquid	**59.36**	K_e 23.67% and 25% DLOM
Growth plan, pre-illiquidity (K_e 20%)	345.67	new machines, new debt, new investors
Growth plan, post-money	259.25	after 25% DLOM
Growth plan, pre-money	**209.25**	less ₹50 cr of new equity

Value creation from the growth plan ≈ 3.5× (209.25 vs 59.36). 30 lakh shares ⟹ **₹864/share**. New investors get ~0.06 cr shares.

*Round 2 mechanics: new debt ₹30 cr, net debt ₹23 cr, equity 112.71 + 50; relevered net β 0.575 → CAPM K_e 12.3%, but the case uses **20%** — new investors may be diversified, but K_e is a negative function of SIZE (Fama-French 1993). WACC = 17.52%.*

IPO proceeds — three treatments

> taken out by owners → **ignore** · used to repay debt → **changes the debt ratio, so the cost of capital and firm value** · held as cash for planned reinvestment → **add to DCF value**

> His closing line: sell to someone who is **well-diversified** (or whose investors are) **and who does not think too highly of you as a person.**

Real options — Laura Martin

*May 1999 · Cox Communications trades at **$37.50** · the sector values on EBITDA multiples · Martin: "they don't ask the right question"*

Method	Value	Note
ROIC target price	$50.00	regression of adj. EV/avg IC on ROIC, **R^2 = 70%**; target multiple 1.594
EBITDA multiple	$50.00	trades 13.3×; target implies **20.9×** — translation, not evidence
DCF	**$54.29**	10 yrs, terminal 13× EBITDA, WACC 9.3% (K_e = 5.12 + 1.07×4.98 = 10.5%, K_d after-tax 3.65%, D/cap 18%, t 39%)
Private market value	≈$51	control premiums on non-public assets

The stealth tier

750 MHz plant: 550 analog + 98 digital ⟹ **102 MHz dark = 17 channels.**
Her objection: "companies are being hit for **100% of the capital spending** but we're only counting the **visible revenue streams**."

```
S = $2,500 per home passed ÷ 108 lit channels    = $23.15
K = opportunity cost: $36/mo = $432/yr; 61% penetration
    = $263/home ÷ 108 = $2.44; at 50% margin      = $1.22
σ = implied vol from a 1-MONTH ATM call on Cox   = 50%
T = life of the cable plant                       = 10 yrs
r = US Treasuries                                 ≈ 5%
→ option value $22.45 × 17 = $381.65 − $40 capex = $341.65 (+14%)
```

Criticise: underlying not traded · 1-month LEVERED EQUITY vol for a 10-yr ASSET option · strike is a judgement · **double-counting if the DCF's TV already grows** · 17 options ≠ 17 × one option.

A real option is worth valuing only if: flexibility + uncertainty + exclusivity. All three.

AI in valuation

> **Valuation theory: unchanged.** DCF, multiples, cost of capital, terminal value.
> **What changes: how the inputs get built and checked.**

Prompts → agents. 2023: single-turn, every step needs a new instruction. Now: the model plans, calls tools and data sources, checks its own output, continues. Workflow: define objective → source data → screen peers → normalise → scenarios → build & validate → output.

Where it helps: peer screening by NLP + clustering (beyond static SIC/GICS); XBRL → Excel; scenarios as driver values.

> **The mistake:** letting it produce a number in prose.
> **His fix:** "Force the model to write the Python code and show the computations for FCF."
> **His disclosure rule:** "write the prompts used, the response obtained, and how the same has been used."
> *Same for equity research reports: use them, give references, never follow the recommendations — mostly biased. Useful for industry information.*

Driver-based revenue — don't grow last year's number

Anthropic: billed **token volume × realised blended price**. 2026 revenue $42,560m — API & cloud 72.8%, Claude Code 11.3%, enterprise 9.3%, consumer 4.2%, other 2.3%. Scenarios: bear 2,600 tn tokens @$8.75/MTok · base 3,100 @$10.00 · bull 3,600 @$10.50.

Maruti: domestic volume × blended realisation; SUV +44.6%, premium +28%; royalty to Suzuki 5–6% of net sales.

> **Which denominator?** $965bn valuation ÷ Q1 annualised $19.2bn = **50x** · H1 annualised $31bn = **31x** · run-rate $47bn = **20x**.
> Run-rate annualises the LATEST period — it is not trailing revenue. Quoting a multiple without the basis is meaningless.

Practical valuation + M&A

When is a valuation legally required?

Income-tax: *shares at premium (**angel tax**), transfer (gift tax), indirect transfer of an Indian asset (**Vodafone**, §9(1)(i) retrospective), slump sale ·* ***Companies Act:*** *preferential allotment, buyback, merger/demerger ·* ***SEBI:*** *open offer, preferential allotment ·* ***FEMA:*** *ODI, R↔NR transfers ·* ***IBC:*** *liquidation ·* ***Ind AS:*** *PPA, options, financial instruments*

Authorised: Registered Valuer (IBBI), CA, Merchant Banker, Chartered Engineer.

Three IVS approaches

Market *— CCM, CTM, PoRI ·* ***Income*** *— DCF, capitalised earnings, excess earnings, relief-from-royalty, with-and-without ·* ***Cost*** *— summation, replacement, reproduction*

> $K_e = R_f + \beta(R_m - R_f) + \text{CSRP}$ CSRP capped ≈ 4%, MRP 6–7%
>
> *Debt/capital, in order of preference: market values → target → industry → book*
>
> *TV: Gordon ·* ***H-model*** *(growth fades — use when capex is heavy) · exit multiple*

Deal patterns

- ***Packaging:*** *no DCF (seller exiting), no comps → offers at 5×. Valued on* ***NAV + goodwill of 2 yrs' PAT*** *→ seller got 8×.*
- ***Die-casting:*** *#1 with 70% share, ₹70 cr EBITDA — hard to sell.* ***Sunset industry*** *⟹ low visibility, low TV, low multiples; ₹450 cr surplus real estate needs a demerger.*
- ***Filtration:*** *₹100 cr+ trapped cash — buyer takes it as* ***dividend ~36%****, seller as* ***capital gain ~15%****. Structure, not argument, closes the gap.*
- ***Staffing auction:*** *same company → 150 / 250 / 250 / 300 / 450.* ***Value is a property of the company AND the buyer.***

> *M&A construct: "EV/EBITDA of x on* ***sustainable EBITDA****,* ***cash-free debt-free****, adjusted for* ***normalised working capital****."*
>
> *Primary = capital creation · Secondary = wealth transfer,* ***15–20% discount****.*
>
> ***Never substitute EPS accretion for economic value creation.***

Quant toolkit + the night before

★ *interim*

$r_{i,t} = \alpha + \beta\, r_{m,t} + \varepsilon$ $\beta = \rho(\sigma_i/\sigma_m)$ $R^2 = \rho^2$

$\sigma_i^2 = \beta^2\sigma_m^2 + \sigma_\varepsilon^2$ **Total beta = $\beta \div \sqrt{R^2}$**

t = coefficient ÷ SE $df = n - k$ threshold $|t| \approx 2$

Test β against 1, not 0. *β = 1.28, SE = 0.19:*
*vs 0 → t = 6.74 (meaningless) · **vs 1 → t = 0.28/0.19 = 1.47 < 2 ⟹ cannot reject β = 1***
95% CI = 1.28 ± 2(0.19) = [0.90, 1.66] ⟹ K_e from 13.1% to 18.4%.

Small samples: *10 comps, 4 parameters ⟹ df = 6 ⟹ critical t = **2.45**, not 2. Report ADJUSTED R^2. And the multiples regression is circular — it assumes the sector is correctly priced.*

Trap sheet — top ten

1. *Adjusting things that need no adjustment (6 of 7 were "not needed")*
2. *Growth with no reinvestment*
3. *Dropping the (1+g), or discounting TV by N+1 periods*
4. *After-tax K_d with CCF*
5. *FCFE at WACC*
6. *Growing FCFE with debt fixed in ₹*
7. *Forgetting the borrowed cash stays in the firm*
8. *Coupon as the cost of debt · book-value weights*
9. *Short-term borrowings inside working capital*
10. *Testing β against 0; trusting a coefficient without its t-stat*

The one habit worth more than any formula: *when handed a set of numbers, check whether they are mutually possible before computing. NI = 0 with positive g · terminal g 10% with a 4× multiple · a 17% IRR project that destroys value · a P/E that reveals ROE at zero growth.*

He is not testing whether you can calculate.
He is testing whether you can tell when a set of assumptions cannot all be true at once.